



GROUP ASSIGNMENT

TECHNOLOGY PARK MALAYSIA

CT042-3-1-IDB

INTRODUCTION TO DATABASES

APU1F2103-APD1F2103-CS-CYB

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WEIGHTAGE: 50%

INSTRUCTIONS TO CANDIDATES:

- 1 Submit your assignment at the administrative counter.**
- 2 Students are advised to underpin their answers with the use of references (cited using the Harvard Name System of Referencing).**
- 3 Late submission will be awarded zero (0) unless Extenuating Circumstances (EC) are upheld.**
- 4 Cases of plagiarism will be penalized.**
- 5 The assignment should be bound in an appropriate style (comb bound or stapled).**
- 6 Where the assignment should be submitted in both hardcopy and softcopy, the softcopy of the written assignment and source code (where appropriate) should be on a CD in an envelope / CD cover and attached to the hardcopy.**
- 7 You must obtain 50% overall to pass this module.**

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ABSTRACT

According to FierceOne Sdn's specifications, this project creates a database system for the BingBing Legend Mobile Game. The project begins by describing the drawbacks of a file-based system in general and for BingBing Legend in particular, as well as how a database system is a more efficient option. The project then identifies the system's business rules, creates nomalized databases, and displays the data in an Entity Relationship. Overall, this paper will outline on how to create a database system for the BingBang Legend Mobile Game.

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1. INTRODUCTION

FierceOne Sdn Bhd. wants to create a new mobile app called BingBing Legend, which is a Multi-Player Online Battle Arena game. The company hired our team consisting of technology professionals to come up with the database system design and implementation of their upcoming game.

To accomplish this, we came up with a concept and gather additional information on both file-based and database management systems, including disadvantages, benefits, an entity relationship diagram, and how they plan to reduce data redundancy through normalization.

Data can help the game producers plan their next move, while further analysis of the data can be used to tailor the game to the specific demands of each customer. Data also aids in assessing each game's current performance, as games are designed to be improved in the hopes of defeating opponents and progressing to the next level. Furthermore, storing data in a database aids the gaming organization in recognizing and identifying their most valuable gamers by allowing professionals to analyze the data and get insights. (ActiveWizards, n.d.).

The gaming business is notorious for frequent updates. Keeping up with changes and trends are critical in order to keep players engaged in the game, as user interest determines whether or not the game will be lucrative. Customers' choices and decisions allow developers to make educated estimates. A solid database gives the game an advantage and a lot of possibilities for machine learning, allowing developers to discover patterns and trends and even make predictions about the future based on the history. (ActiveWizards, n.d.).

Our team intends to make the concept as smooth and simple as possible for users, not only in terms of ease of use, but also in terms of efficiency through the use of normalization. Our team will take in count all of the data that Bing Bing Mobile Legends will need to store in the database, as well as every small and crucial element that will be required to make the database as efficient and functional as possible.

2. DISADVANTAGES OF FILE-BASED SYSTEM

The operating system's mechanism and data structure for specifying the files on a disc or partition is known as a file system. It also refers to the disc or partition used to store the files, as well as the type of file system employed. A file management system, or file system for short, is the software structure in an operating system that is responsible for organizing and storing file information (Abhishek, 2020). The software involved with file management, the files being managed, and the data structures required to accomplish file management make up a file system. The file system is the system that organizes and allocates file memory space, is in charge of file storage, and protects and retrieves the deposited data from a system viewpoint. It oversees generating files for users, depositing, reading, changing, and dumping files, managing file access, and revoking files when they are no longer in use, among other things. A computer-based file-based system offers several advantages over a manual record-keeping method, but it also has several disadvantages. The following are the major drawbacks of file-based computer systems:

2.1. Data redundancy – it is the first drawback. The existence of redundant data in separate data files is referred to as data redundancy. Data is structured into distinct data files in a document management system, each having its architecture, and actions on the data are accessed by file name (Watt, 2014). The data files have no link to one another; they are application oriented. Each program has and utilizes its own data file, and each data file unavoidably contains numerous data duplicates. In database systems that manage huge volumes of shared data in a database format, the danger of data redundancy is increased. The in-game ID cannot be used to identify the player profile in the game if it is not a unique sequence. When looking for specific data, it may provide erroneous or nonsensical results, causing the process to fail to owe to data duplication.

2.2. Data Integrity – The issue of data integrity is another disadvantage of using a file-based system. The quality and consistency of data in a database are referred to as "data integrity." There are a few things to consider while dealing with this issue. The data values must meet certain consistency requirements specified in the application. Consequently, all the correct and limiting requirements must be met by the data in the database. In file processing systems, poor data integrity is prevalent, necessitating the programmer's incorporation of these restrictions via code (Pearlman, 2019). Many strategies are employed to establish data integrity and non-repudiation in the gaming environment, for example, the requirement to safeguard and trust the information transmitted between gaming devices and their hosts grows as more gaming devices and systems use the Internet and other communication methods to extend their capabilities.

2.3. Data Inconsistency – Data inconsistency is the file processing system's third drawback. Different files may include different information about the same thing or person, which is known as data inconsistency (Prashant, 2019). Redundancy, in effect, leads to inconsistency. When the same data is kept in various locations, inconsistencies might develop. This occurs because files containing duplicate data are not correctly listed. Different copies of the same data in a file processing system may have different values. The data in this system is inconsistent, which means that if one data item must be altered, all the files holding that data must be adjusted as well. As a result, there's a chance that the data values will be out of date.

If the player's username is updated in the data system, the gamer's username should be changed in all departments where the gamer is involved.

2.4 Atomicity Problems – The Atomicity Problem is the fourth disadvantage. In a file processing system, atomicity is difficult to accomplish. Data atomicity refers to the fact that data or records are either inputted in their whole or not at all. Atomicity is required for data values to be preserved, and it indicates that information is either totally input or completely cancelled. Any system can fail at any time, and one would expect the data to be in a consistent condition at that time. If any insertion, update, or deletion in the file system fails, there is no way to return to the prior state. Consider our game's scenario: the game's data must be recorded into a report file, but the file is closed without being saved once the new score is entered. This indicates that the full transaction is not completed. The only thing that was done was to collect game data, not to add to it. In this scenario, the overall score calculated is incorrect. Atomicity denotes the completion of the full transaction or the absence of any transaction (Rahul, 2019). Any transaction that is only partially completed will result in inaccurate data in the system.

Hence these kinds of integrity checks are not easy in file systems. Data is saved as files on external storage and handled by the operating system during the file system stage. Users can access files using a user-friendly interface provided by the operating system. The file's logical structure is separated from its physical structure, and the program and data are separated, allowing the data to function independently of the program. The user's program and data can be saved on the external memory independently, and each application can share a set of data, allowing for file-based data sharing. However, there is a lot of data redundancy because data is still organized programmatically. Furthermore, the logical structure of the data is difficult to modify and expand, and even minor changes to the logical structure of the data have an impact on the application. Because the files are independent of one another, they do not represent the links that exist in the actual world, and the operating system is not responsible for keeping track of file relationships. If there is a content link between files, the program is the only one who can manage it.

3. ADVANTAGES OF DATABASE, DBMS AND ITS FUNCTIONS

Database Management System (DBMS) : It is a type of large-scale database manipulation and management software that is used to create, operate, and maintain databases. It administers and controls the database in a centralized manner to ensure the database's security and integrity. The database management system allows users to access data in the database, and database administrators can use it to do database maintenance. It has several features that allow numerous applications and users to build, change, and query databases in various ways at the same time or at separate times. It enables users to quickly create and manage data, maintains data security and integrity, and govern and restore data concurrently across several users. The database management system provides several advantages as a result of its centralized management and control, the most notable of which are the following:

3.1. Database management system's security – In a database system, it is easy to set access limits so that only authorized users have access to the data. Data is protected from risks such as identity theft, data leakage, and data abuse since each user has separate access privileges. The Database Administrator ensures that the appropriate access is granted, including the appropriate authorization model for database management system access and checks prior to accessing sensitive data. When accessing the database, the Database Administrator can define usernames and passwords to identify authorized users. To accommodate diverse types of data and procedures, several levels of security measures might be established. Varied authorized users have different levels of access to the database's various sorts of data.

3.2. Data Sharing – The second benefit of employing a database system is that it enables a shared database to be created. Data sharing refers to the ability for all users to access the database's data at the same time, as well as the numerous methods in which users may interact with the database and share data. Because the database system is a centralized repository for all of the organization's data, it may be accessed by anyone with permission. The data in the database may be shared by existing apps. Furthermore, without reinventing all data needs, new applications can rely on data that already exists in the database and only contribute data that is not recorded. Therefore, more users will be able to share more information. Since game data may be shared, the player's basic information records, such as personal information and detailed overall information about the game record, can be monitored and seen after the game has ended.

3.3. Data Redundancy – Data redundancy may be reduced to a reasonable extent in database systems. When creating a database as a unified, logically connected, centralized structure, diverse user groups' perspectives are incorporated. Because of the centralized database and the Database Administrator's centralized control of data, unnecessary data duplication is eliminated. In an ideal world, each main piece of data is stored in only one place in the database, reducing total data storage needs and eliminating the need to keep track of huge volumes of extra data (Alter, 2014). Of course, we're not suggesting that every redundancy should or must be avoided; in certain cases, genuine transactional or technical reasons may necessitate the maintenance of numerous copies of the same data. These redundancies must be carefully managed in a database system. The database, as opposed to a file system, allows data sharing, eliminating the need for users to build their own application files. There is a

significant reduction in duplicate data, which decreases data redundancy and preserves data consistency.

3.4. Data consistency – The greater data consistency is the fourth point of strength. When data is modified on one site without propagating the altered value to its copies on other sites, inconsistency occurs, resulting in inaccurate or conflicting data being presented to users (Bowman, 2013). As a result, removing and controlling duplication also eliminates and controls the possibility of data inconsistency. When data is submitted to a database system, inconsistency is typically minimized since changes to any one piece of data in the database are immediately replicated to all copies of it. This is known as propagating updates. The danger of inconsistency is reduced by ensuring data consistency. If a data item is only saved once in the database, each modification to that value is only performed once, and the new value is available to all users immediately. If the system is aware that the data item is saved more than once, it will verify that all copies of the item are consistent.

The ability of end-users and programmers to access and use the same data is perhaps the most essential benefit of a database management system. More significantly, data integrity is not jeopardized. A database management system (DBMS) ensures data independence by providing flexible access to data and ensuring access without requiring the user to know where the data is stored (Watt, 2014). Users no longer have to be concerned about data structure changes because of data independence. A database management system can also improve data integrity and security. Data is not readily compromised since several degrees of access to the same data can be granted. Because most DBMSs also indicate that data is stored in a different physical place than the user, data security is further reinforced. In addition, duties like data backup and recovery are part of the basic functionality. When data is accessed and updated, this guarantees that it is not compromised. It ensures a uniform data management procedure in general.

4. LIST OF BUSINESS RULES

1. Player has name, email address, username, online status, number of diamonds the player has, total battle points, global level and identified by player ID.
2. Hero has name, role of the hero, hero's speciality, current level of the hero, cost of the hero and identified by hero ID.
3. Player Hero has only level and identified by player ID.
4. Game has game mode, game result, total game time, match data, mode id and identified by game ID.
5. Skin has skin cost, skin name, skin ID and identified by hero ID.
6. There will be a starter hero for new players that can be unlocked for free.
7. Regardless of the game mode, the choice of heroes by every player must be unique in a team but not among opponent's team.
8. If a player quits in the middle of a game, the game will not be recorded for that player and the remaining four players in the same group will proceed with the game.
9. When a game is over, each player's game history will record the results of the previous game, the detailed overall information about the players and the MVP of the game.
10. When a game is over, each player on the winning team will receive battle points and diamonds as their reward. The losing team will receive fewer battle points, but no diamond.
11. "Each game has 3 types of modes which are Brawl mode, Human vs AI mode, and Classic Mode.
12. Each game must have one arena.
13. Each game mode must have one unique ID and Name.
14. Each game needs 2 teams to start the game.
15. Each team must have five members called active-players.
16. To purchase a skin for a certain hero, players must own at least one of them.
17. Battle points or diamonds can be used to purchase heroes and hero skins.

18. Each active player in each game must select a hero to represent him in battle.
19. For use in the game's battles, one skin from each of the heroes involved will be chosen.
20. Players can keep track of each other's fundamental information, such as their profile, prior game outcomes, and most-used heroes, as well as detailed general statistics about the player's previous games.

5. NORMALIZATION (FROM UNF → 3NF)

5.1. Normalization:

Database normalization is a database schema design approach that involves modifying an existing schema to reduce data redundancy and reliance. This includes generating tables and defining relationships between them according to rules aimed to secure data while also allowing the database to be more agile by removing duplication and inconsistent dependencies.

5.2 Unnormalized form (UNF)

A UNF table is a table that contains one or more repeating groups. Here is an example of UNF table:

Player_ID	Username	Global Rank	Hero_ID	Hero_Name	Role	Hero_level
P001	Alice	11	H002	Oscar	Tank	3
			H006	Whiskey	Mage	5
P002	Bob	5	H001	Juliett	Archer	5
P003	Emma	30	H005	Sierra	Archer	8
			H006	Whiskey	Mage	1
P004	Romeo	43	H001	Juliett	Archer	7
			H003	Mike	Fighter	7
			H004	Victor	Assassin	9
			H007	Lima	Fighter	3

Fig:1. UNF

Player_Hero (Player_ID, Username, Global_Rank, (Hero_ID, Hero_Name, Role, Hero_level))

5.3. First Normalized Form (1NF)

To normalize the table from UNF into 1NF, the data should be presented in tabular format to eliminate nulls by making sure that each cell has a single value.

Next is to identify a primary key. A primary key must uniquely identify attribute value and the new key must be composed. In this case, the primary key would be the composite of Player_ID and Hero_ID.

<u>Player_ID</u>	Username	Global_Rank	<u>Hero_ID</u>	Hero_Name	Role	Hero_level
P001	Alice	11	H002	Oscar	Tank	3
P001	Alice	11	H006	Whiskey	Mage	5
P002	Bob	5	H001	Juliett	Archer	5
P003	Emma	30	H005	Sierra	Archer	8
P003	Emma	30	H006	Whiskey	Mage	1
P004	Romeo	43	H001	Juliett	Archer	7
P004	Romeo	43	H003	Mike	Fighter	7
P004	Romeo	43	H004	Victor	Assassin	9
P004	Romeo	43	H007	Lima	Fighter	3

Fig:2. 1NF

Player_Hero (Player_ID, Username, Global_Rank, Hero_ID, Hero_Name, Role, Hero_level)

Lastly, we need to list out all the functional dependencies, which included partial dependency and transitive dependency. The functional dependency is as follow:

Functional Dependencies:

Player_ID → Username, Global_Rank (partial dependency)

Hero_ID → Hero_Name, Role (partial dependency)

Username → Global_Rank (transitive dependency)

5.4. Second Normalization Form (2NF)

A relation must be in first normal form(1NF) and contain no partial dependencies to be in second normal form. Therefore, we need to move those partial dependencies that were determined in 1NF and place them into new tables. Most anomalies have been removed at this point.

In this example, the username and global rank can be determined by Player_ID, meanwhile hero names and their roles can be determined by a Hero_ID. Hence, we have to add 2 new tables for them to eliminate partial dependencies.

<u>Player ID</u>	<u>Hero ID</u>	Hero_level
P001	H002	3
P001	H006	5
P002	H001	5
P003	H005	8
P003	H006	1
P004	H001	7
P004	H003	7
P004	H004	9
P004	H007	3

<u>Player ID</u>	Username	Global Rank
P001	Alice	11
P002	Bob	5
P003	Emma	30
P004	Romeo	43

<u>Hero ID</u>	Hero_Name	Role
H001	Juliatt	Archer
H002	Oscar	Tank
H003	Mike	Fighter
H004	Victor	Assassin
H005	Sierra	Archer
H006	Whiskey	Mage
H007	Lima	Fighter

Fig:3. 2NF

Player_Hero (Player ID, Hero ID, Hero_level)

Player (Player ID, Username, Global_Rank)

Hero (Hero ID, Hero_Name, Role)

5.5. Third Normalized Form (3NF)

A relation is in third normal form if there is no transitive dependency for non-prime attributes and the relation is in second normal form. To eliminate all the transitive dependencies, we need to move those transitive dependencies that were determined in 1NF and place them into new tables. 3NF is considered the highest level necessary for most applications.

In this case, since the username that the player provides has to be unique, therefore username has the ability to determine the user's global rank and it will be considered as a transitive dependency.

<u>Player ID</u>	<u>Hero ID</u>	Hero_level
P001	H002	3
P001	H006	5
P002	H001	5
P003	H005	8
P003	H006	1
P004	H001	7
P004	H003	7
P004	H004	9
P004	H007	3

<u>Player ID</u>	Username
P001	Alice
P002	Bob
P003	Emma
P004	Romeo

<u>Hero ID</u>	Hero_Name	Role
H001	Juliett	Archer
H002	Oscar	Tank
H003	Mike	Fighter
H004	Victor	Assassin
H005	Sierra	Archer
H006	Whiskey	Mage
H007	Lima	Fighter

Username	Global Rank
Alice	11
Bob	5
Emma	30
Romeo	43

Fig:4. 3NF

Player_Hero (Player ID, Hero ID, Hero_level)

Player (Player ID, Username)

Hero (Hero ID, Hero_Name, Role)

PlayerRank (Username, Global_Rank)

6. ENTITY RELATIONSHIP DIAGRAM (ERD)

6.1. Definition: The relationship between entity sets is portrayed using an Entity relationship diagram. An entity set is a collection of similar entities that may or may not contain attributes. An entity in a database management system is a table or an attribute of a table, hence an ER diagrams show the entire logical structure of a database by displaying relationships between tables and their attributes. (Hoffmann, 2016)

The following are the main components of an ERD:

1. Entities
2. Relationships
3. Attributes

The following are the steps involved in creating an ERD:

1. Identifying the entities.
2. Identifying all interactions between the entities.
3. Identifying the cardinality of relations and analyzing the nature of interactions.
4. Finally creating the ERD.

6.2. Constructing the Database:

We have used Chen's notation to design our ERD because it is simple and easy to navigate and comprehend.

6.3. Tool used:

We chose the darw.io tool software over Microsoft Visio and other diagram software because it is a more efficient and effective program for producing and updating diagrams. It also offers several graphics and features that make programming easy and precise.

6.4 ENTITY RELATIONSHIP DIAGRAM DESIGN

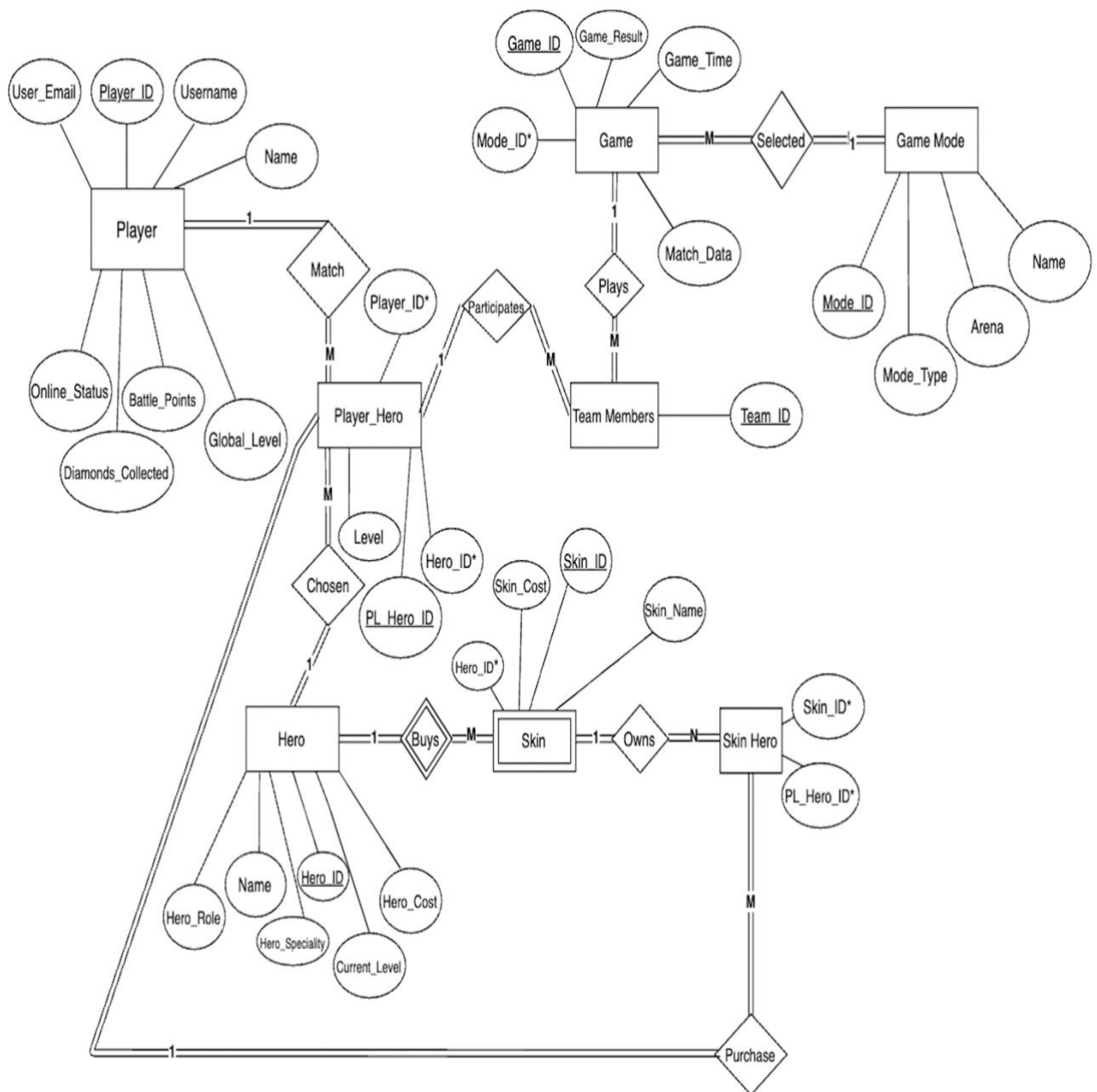


Fig:5. Entity Relationship Diagram

7. CONCLUSION

It is clear that while archaic data storage methods such as file-based systems may have sufficed in the past, with the rapid advancement of technology and information, the transition to Database Management Systems has become more of a necessity than a luxury. And, in the case of the Bing Bing Mobile Legends Game, it's an important element for the entire process, starting from player registration down to the results after a match end.

By setting clear business rules and instructions, the outline of the database has been made and a functioning design was created and implemented. Furthermore, the normalization process serves a wonderful guide for creating a database that is well-designed and has few redundancies. Moreover, the normalization process serves as a wonderful guide for creating a database that is well-designed and has few redundancies. Leading To an unified system where any piece of data has a place and a function and access to any set of data is easier and more efficient, leading to a more productive work environment for staff working on Bing Bing's development, as well as overall customer happiness.

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WORKLOAD MATRIX

1. S M RABEYA AMIN IKRA				
Project Title	INTRODUCTION AND CONCLUSION			
Project Effort	3 Days			
Project status	Complete			
2. GOH WEI SONG				
Project Title	Database and Database Management System			
Project Effort	5 Days			
Project status	Completed			
3. SAW CHIOU MING				
Project Title	UNF	1NF	2NF	3NF
Project Effort	3 Days	2 Days	2 Days	2 Day
Project status	Complete	Complete	Complete	Complete
4. ABDULLAH AL MUKIT				
Project Title	Entity Relationship Diagram			
Project Effort	10 Days			
Project status	Complete			